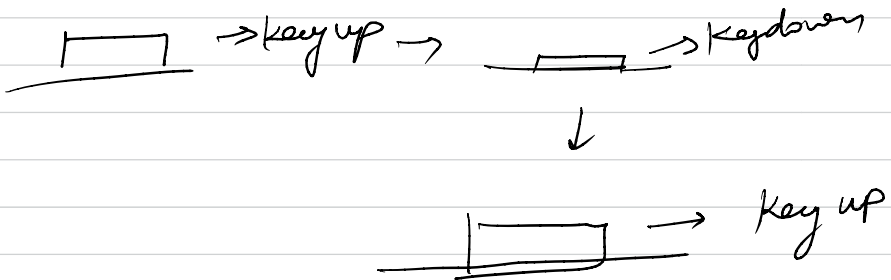



3 Aug, 2024

DOM Continued :-

- ~~append~~ allows string & dom element.
- append child → only allows dom element (not string)
- remove()
- removeChild() → khud se pedo

Events :-



add event listener me "on" mat likho

only mouse and keyboard events pado from W3School



event handler function

e.target.value → only for i/p.

vanilla.js → pure JS ko kehte hai

e.which → to find kounsi key user ne press kari.

DOM continued :-

5 Aug, 2024

[Shreyans online video → 15]

`remove()` → syntax → `element.remove()`;

`removeChild()` → syntax → `parentNode.removeChild(childNode)`;

HTML code

```
<ul class="movies">
  <li>AB</li>
  <li id="li-movies">CD</li>
  <li>EF</li>
</ul>
```

Before → AB

CD

EF

```
const li = document.querySelector('#li-movies');
li.remove();
```

Now → AB

EF

↘ JS code

let's see how to use `removeChild()`

HTML code :-

```
<ul class="movies">
  <li>AB</li>
  <li id="li-movies">CD</li>
  <li>EF</li>
</ul>
```

JS code :-

```
const liClass =
document.querySelector('.movies');

const liId =
document.querySelector('#li-
movies');

liClass.removeChild(liId);
```

Before: AB

CD

EF

After: AB

EF

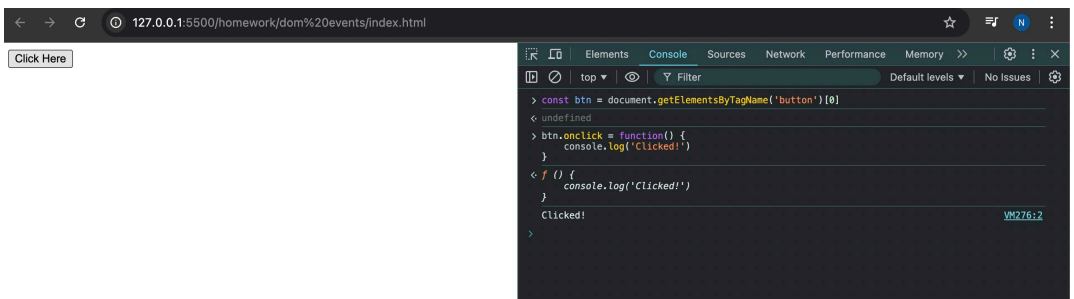
you can also write $\rightarrow$ `liClass.removeChild`
(`'ul li : second-child'`);

#DOM Events :- user website par jobhi activity perform kar raha hai $\rightarrow$ wo event hai
For example: Typing, mouse click, hover, etc.

Aksar hamare saare events "on" se shuru hote hai.
`onclick`
`ondrag`
`ondrop`, etc.

① Let's say tum ek button par click karo, then console mei aage "Clicked". Note! you have to change / create function inside `onclick` property of button.

Ans



The screenshot shows a web browser window with the address bar displaying `127.0.0.1:5500/homework/dom%20events/index.html`. A button labeled "Click Here" is visible. The browser's developer console is open, showing the following code and output:

```
> const btn = document.getElementsByTagName('button')[0]
< undefined
> btn.onclick = function() {
  console.log('Clicked!')
}
< f() {
  console.log('Clicked!')
}
Clicked! VM276:2
```

Hamne `onclick` ki definition / functionality change kar di. Jaise hi maine Click Here par click kiya, console mei $\rightarrow$ "Clicked" aa gaya.

usually onclick is null



```
Elements Console Sources Network Performance Memory >>
top Filter Default levels No Issues
Live reload enabled. index.html:40
> const btnn = document.getElementsByTagName('button')[0]
< undefined
> btnn.onclick
< null
> btnn
< <button>Click Here</button>
```

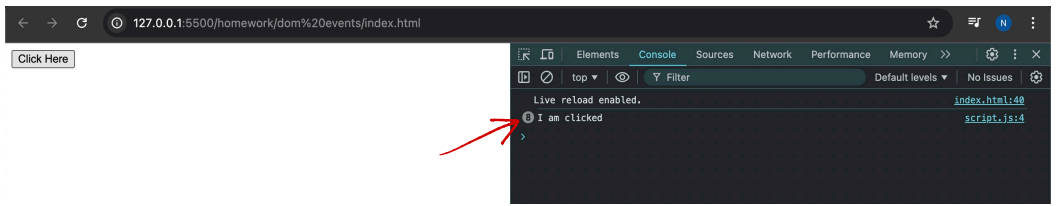
you can also do → `console.dir(btnn)` to find onclick value amongst all other events.

```
Elements Console Sources Network Performance Memory >>
top Filter Default levels No Issues
oncancel: null
oncanplay: null
oncanplaythrough: null
onchange: null
onclick: null
onclose: null
oncontentvisibilityautostatechange: null
oncontextlost: null
oncontextmenu: null
oncontextrestored: null
oncopy: null
oncuechange: null
oncut: null
ondblclick: null
ondrag: null
ondragend: null
ondragenter: null
ondragleave: null
ondragover: null
ondragstart: ondragleave
ondrop: null
```

usually hum add event listener use krte hai.
HTML JS

`<button> Click Here
</button>`

```
const btn =  
document.getElementsByTagName('button')[0]  
  
btn.addEventListener('click', ()=>{  
  console.log('I am clicked');  
})
```



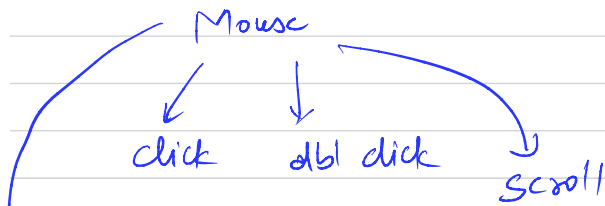
syntax $\rightarrow$ `element.addEventListener('event', function)`

Note: addEventListener by default
ek object add kar deta
hai inside function

yaha hum
"on" use
nahi karenge

event hone
par jo function
run hoga vo
yaha aayega.

Reference:- Shreyans video 16



Mouse ke aur bhi events hai which you
can find/ study from w3 school.

Hum sirf mouse & keyboard event padenge.

syntax $\rightarrow$ `element.addEventListener('event', function)`

Note: `addEventListener` by default
ek object provide
karata hai

yaha hum
"on" use
nahi karenge

event hone
par jo function
run hoga vo
yaha aayega.

which object?

Is object ko event object kehte hai

`element.addEventListener('click', (e) => {`

`});`

or write

you can call it `e`, `event`, `ev`, etc. `e` ko console
karake dekhte hai ki `e` kya chiz hai. check code $\searrow$

```
homework > dom events > index.html > html > body
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport"
6     content="width=device-width, initial-scale=1.0">
7   <title>Document</title>
8   <style>
9     h1{
10       border: 2px solid salmon;
11     }
12 </style>
13 </head>
14 <body>
15   <!-- <button>Click Here</button> -->
16
17   <h1>Learning DOM</h1>
18
19   <script src="./script.js"></script>
20 </body>
21 </html>
```

```
homework > dom events > JS script.js > ...
1 // const btn = document.getElementsByTagName('button')
2 // [0]
3 // btn.addEventListener('click', ()=>{
4 //   console.log('I am clicked');
5 // })
6
7 const h1 = document.querySelector('h1');
8
9 h1.addEventListener('mouseenter', (e)=>{
10   console.log(e);
11 })
```

Learning DOM

jaise hi cursor leke
border ke andar
gaye, waise hi
ye o/p aaya
in console.

```
MouseEvent script.js:10
MouseEvent {isTrusted: true, screenX: 203, screenY: 179, clientX: 203, clientY: 61, ...} script.js:10
MouseEvent {isTrusted: true, screenX: 244, screenY: 179, clientX: 244, clientY: 61, ...} script.js:10
MouseEvent {isTrusted: true, screenX: 235, screenY: 180, clientX: 235, clientY: 62, ...} script.js:10
MouseEvent {isTrusted: true, screenX: 234, screenY: 180, clientX: 234, clientY: 62, ...} script.js:10
MouseEvent {isTrusted: true, screenX: 613, screenY: 162, clientX: 613, clientY: 44, ...} script.js:10
MouseEvent {isTrusted: true, screenX: 306, screenY: 180, clientX: 306, clientY: 62, ...} script.js:10
MouseEvent {isTrusted: true, screenX: 305, screenY: 177, clientX: 305, clientY: 59, ...} script.js:10
MouseEvent {isTrusted: true, screenX: 301, screenY: 173, clientX: 301, clientY: 55, ...} script.js:10
MouseEvent {isTrusted: true, screenX: 295, screenY: 140, clientX: 295, clientY: 22, ...}
```

jab aap ye Mouse Event khologe, then
aapko isme kafi sare kya milegi.

you will also find target = "h1"

type = "mouseenter", etc.

screen X, screen Y → coordinates on screen.

Note → aapko kiye gaye event se related
sabhi info mil jayegi yaha par.

Learning DOM

```
Elements Console Sources Network Performance Memory Application Security Lighthouse Recorder
top Filter
▶ fromElement: html
  layerX: 203
  layerY: 61
  metaKey: false
  movementX: 0
  movementY: 0
  offsetX: 193
  offsetY: 38
  pageX: 203
  pageY: 61
▶ relatedTarget: html
  returnValue: true
  screenX: 203
  screenY: 179
  shiftKey: false
▶ sourceCapabilities: InputDeviceCapabilities {firesTouchEvents: false}
▶ srcElement: h1
▶ target: h1
  timeStamp: 10364.699999988079
  toElement: h1
  type: "mouseenter"
▶ view: Window {window: Window, self: Window, document: document, name: '', location: Location, ...}
  which: 0
  x: 203
  y: 61
```

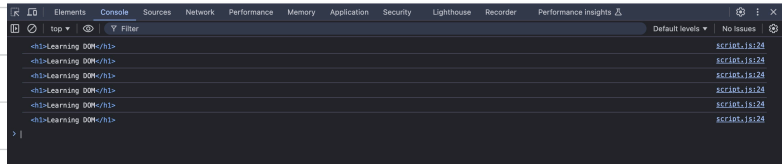
```
homework > dom events > index.html > html > body
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport"
6     content="width=device-width, initial-scale=1.0">
7   <title>Document</title>
8   <style>
9     h1{
10       border: 2px solid salmon;
11     }
12 </style>
13 </head>
14 <body>
15   <!-- <button>Click Here</button> -->
16
17   <h1>Learning DOM</h1>
18
19   <script src="./script.js"></script>
20 </body>
21 </html>
```

```
homework > dom events > JS script.js > ...
1 // const btn = document.getElementsByTagName('button')
2 // [0]
3 // btn.addEventListener('click', ()=>{
4 //   console.log('I am clicked');
5 // })
6
7 const h1 = document.querySelector('h1');
8
9 h1.addEventListener('mouseenter', (e)=>{
10   console.log(e);
11 })
```

console.log(e.target);

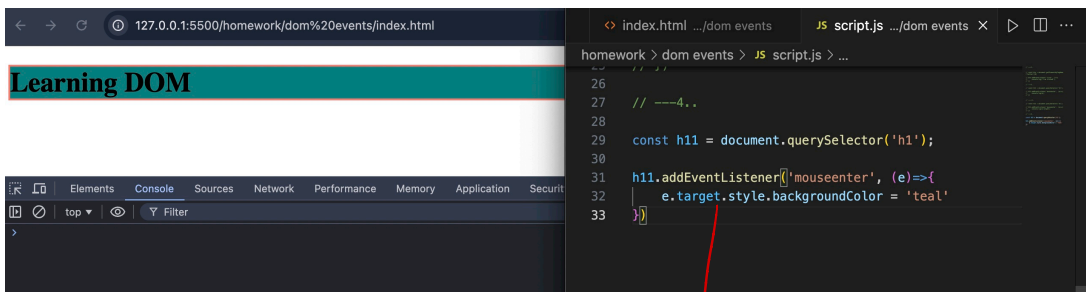
*ye likhne per o/p
aayega.*

Learning DOM



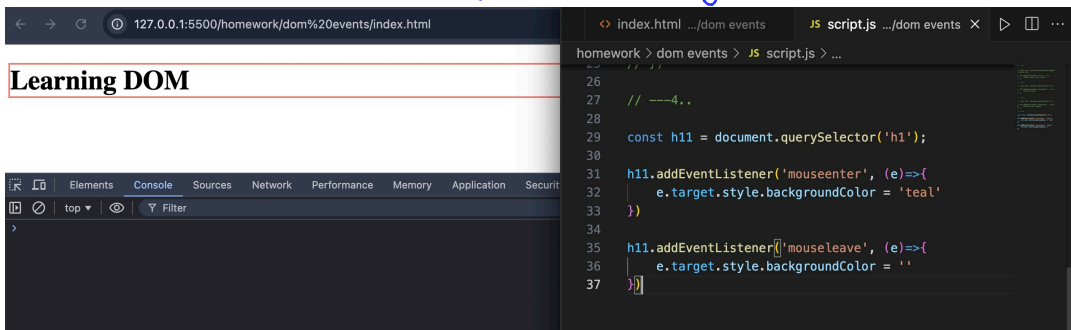
→ jaise hi border ke andar gaya, waise hi h1 ki value aa gayi.

Agar mujhe border ke andar jaane par bgcolor change karna ho, tab kya kare?



style change kar diya

But isme ek problem hai, mai jaise hi cursor border se bahar laaunga color change ~~ko~~ nahi hoga. Ab kya kare?



Mum ek aur separate event add kar diya
jab mouse bahar aaye border se.

mouse down → jab mouse click kiya

mouse up → jab mouse click release
kiya



Same with keyboard Key down → key
pressed

Key up → key
released

Note:- Mouse click karne par do events ho
sakte hai → press & release

Keyboard events :-

key down

key up


```
continued  index.html .../dom events X JS script.js .../dom ev ... JS script.js X
homework > dom events > index.html > html > body > input
2 <html lang="en">
13
14 <body>
15
16 <!-- <button>Click Here</button> -->
17
18 <h1>Learning DOM</h1>
19
20 <input type="text" placeholder="type here">|
21
22 <script src="/script.js"></script>
23
24 </body>
25 </html>

35 // h1.addEventListener('mouseleave', (e)={
36 //   e.target.style.backgroundColor = ''
37 // })
38
39
40 // ---5..
41
42 const inp = document.querySelector('input')
43
44 inp.addEventListener('keypress', (e)=>{
45   console.log(e);
46 })
```

Maine Keyboard se jo bhi type kiya uske related info o/p mei aa jayegi.

Learning DOM

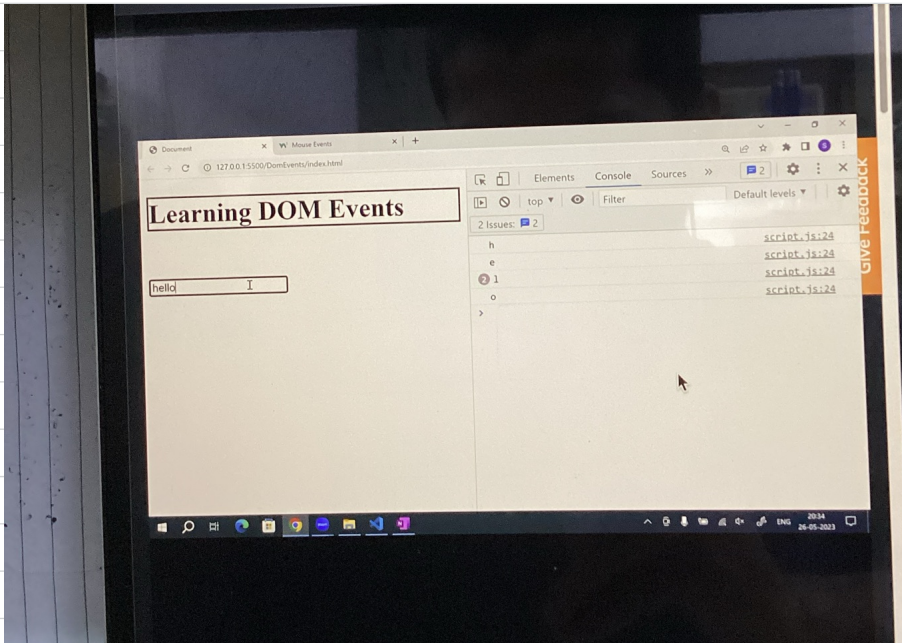
t

```
Elements Console Sources Network Performance >> 1 Issue: 1
top Filter Default levels 1 Issue: 1
{ bubbles: true,
  cancelBubble: false,
  cancelable: true,
  charCode: 116,
  code: "KeyT",
  composed: true,
  ctrlKey: false,
  currentTarget: null,
  defaultPrevented: false,
  detail: 0,
  eventPhase: 0,
  isComposing: false,
  key: "t",
  keyCode: 116,
  location: 0,
  metaKey: false,
  repeat: false,
  returnValue: true,
  shiftKey: false,
  sourceCapabilities: InputDeviceCapabilities {firesTouchEvents: false},
  srcElement: input,
  target: input,
  timeStamp: 4831.699999988079,
  type: "keypress",
  view: Window {window: Window, self: Window, document: document, name: '', location: Location, ...},
  which: 116 }
[[Prototype]]: KeyboardEvent
```

maine "t" press kiya tha, wo show ho raha hai.
target was input tag

Agar ye likha instead of console.log(e)
↓
console.log (e. key)

↳ jo jo press karoge wo
console mei show hoga
↓



Learning DOM

ty

Elements Console Sources Network Performance

top Filter

t	script.js:45
t	script.js:49
y	script.js:45
y	script.js:49
p	script.js:45
p	script.js:49
Backspace	script.js:45
Backspace	script.js:49
Meta	script.js:45
Shift	script.js:45

index.html .../dom events JS script.js .../dom events

homework > dom events > JS script.js > inp.addEventListener('keyup') callbac

```
28
29 // const h1 = document.querySelector('h1');
30
31 // h1.addEventListener('mouseenter', (e)=>{
32 //   e.target.style.backgroundColor = 'teal'
33 // })
34
35 // h1.addEventListener('mouseleave', (e)=>{
36 //   e.target.style.backgroundColor = ''
37 // })
38
39
40 // ---5..
41
42 const inp = document.querySelector('input')
43
44 inp.addEventListener('keydown', (e)=>{
45   console.log(e.key);
46 })
47
48 inp.addEventListener('keyup', (e)=>{
49   console.log(e.key);
50 })
```

Hamne ek aur event add kar diya for "keyup"

Ab mai key dabaunga but do s/p denge
console mei. Kyuki key press karne par
do events hote "press" and "release"

let's create a form:-

jaise hi mai submit/button par click karunga,
page reload hoga. This is the behavior of
form. How to avoid it?

use `e.preventDefault()` to prevent default
behavior of the event.

`e.target.value` → you will get the
value which you are
typing.

Q jaise hi mai Submit par click karu,
uske mujhe user name and password
ki value console mei aa ~~ga~~ jaye
jo maine type ki thi. kaise kare?

```
<form action="">
```

```
  <input type="text" placeholder="username" name="username"> l1.  
  <br>  
  <input type="password" placeholder="password" name="pass"> l2  
  <br>  
  <button>Submit</button>
```

```
</form>
```

→ html
code

l1
l2
const userNameInput = document.getElementsByName('username')[0];

const passInput = document.getElementsByName('pass')[0];

const form1 = document.querySelector('form');

form1.addEventListener('submit', (e) => {
 e.preventDefault();
 console.log(userNameInput.value);
 console.log(passInput.value);

userNameInput = '';
 passInput = '';
})

→ JS code

l6, l7 correct
syntax

userNameInput.
value
passInput.value

Agar html ke l1 and l2 mei maine name
attribute add kiya hai: name = "username"
name = "pass"

JS code explanation:-

① l1, l2 mei type kiye hue username, ~~ga~~
password ki value store karayi.

② `l3` is written to prevent default behavior of form which is to prevent reloading when submit is clicked.

③ Submit par click hote hi user dwara type ki gayi value will be visible in ~~op~~ console. (`l4, l5`)

④ `l6, l7` → value clear / empty kardo.

Let's make a calculator :-

1:32:00 → yaha se JS ka code shuru hoga.

① Sabse pehle sabhi button and i/p ko select kar lo.



i/p wali screen

② Ab aapko button par event listener lagana hai. Ab ya to har button par individually lagao ya loop lega do.

Redo → Fir se
calculator kamao
with step by
step execution

(1:45:00)

```

11 const buttons = document.querySelectorAll('button');
12 const screen = document.getElementById('screen');

23 for(let button of buttons){
24     button.addEventListener('click', (e)=>{
15         const buttonText = button.innerText;

16         if(buttonText === 'C'){
17             screen.value = "";
18         } else if(buttonText === 'x'){
19             screen.value = screen.value + '*';
20         } else if(buttonText === '='){
21             try {
22                 screen.value = eval(screen.value);
23             } catch (error) {
24                 screen.value = "Invalid Operator!";
25             }
26         } else{
27             screen.value = screen.value + buttonText;
28         }
29     })
30 }

```

Code Explanation :-

- ① Sabse pehle hamne sabhi button ko select kiya.
- ② Then hamne uss element ko select kiya jaha screen is set as ID. The element is i/p element.
- ③ 23 par hamne loop kyu lagaya?
Agar hum loop nahi lagayenge, then hame individually buttons ko one by one select karna padega. And har button ka code karna padega. It will increase the length of code.
- ④ Jaise hi mera koi bhi button click hoga, uss button par jo bhi text likha hua hai, it will be saved in "buttonText"

l4 par hamne button.innerHTML use kiya hai.
Kyu? Hamne yaha button.value kyu use
nahi kiya?

innerHTML is used to get/set the text
element written inside it.

<h1> Text Msg </h1>
↪ here you can innerText.

l5 mei hamne screen.value kyu use kiya?

Input element par hum jo bhi hum type
karke hai, vo by default value mei jata
hai. → you can do console.dir(screen)
to check the value of "value".

⑤ l5 → agar Clear/C par click hua, then
screen.value empty kardo.

⑥ l6 → agar X par click kiya, then
screen.value mei concatenate kardo
"X" ko

⑦ l7 → "=" click hote hai ~~to~~ screen.value
mei jo bhi mathematical expression
hai. use evaluate kardo.

l8 ⇒ Agar evaluate karne par error aaya

then code "then" ke baad "catch" mei
chalega

~~code~~

③ lg → is for remaining buttons